INFO B302 – Introduction à la démarche scientifique (IDS)

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Modalités

- L’activité d’apprentissage *introduction à la démarche scientifique* se présente sous la forme d’un séminaire.
- Le fil conducteur s’appuie sur les points essentiels de la réalisation du mémoire du master en sciences informatiques
  - Réfléchir
  - Élaborer
  - Communiquer
- Travaux de groupe

Objectifs

- Vous faire découvrir ce qu'est une **démarche scientifique** en vous donnant l'occasion de **mener** une telle démarche ;
- Vous apprendre à réfléchir sur la démarche scientifique ;
- Mettre l'accent sur votre **capacité d'analyse réflexive** en vous incitant à mener une réflexion sur vos propres actions, **développer un esprit critique** et **expliquer et argumenter** vos démarches ;
- Vous préparez à la démarche de réalisation d'un **mémoire de master**.

Évaluation

- Cette activité d'apprentissage demande une **participation active** sous la forme de *déliverables* à remettre tout au long du quadrimestre et qui formeront un **rapport final de *pré-enregistrement*** (*registered report)*.
- Les différents travaux sont à réaliser en **en anglais** par **groupe de 3 étudiants**. La **note finale est individuelle** et consiste en la participation aux différentes activités du cours, l'évaluation du rapport de pré-enregistrement et une défense du travail lors de l'examen.

About the lecturer

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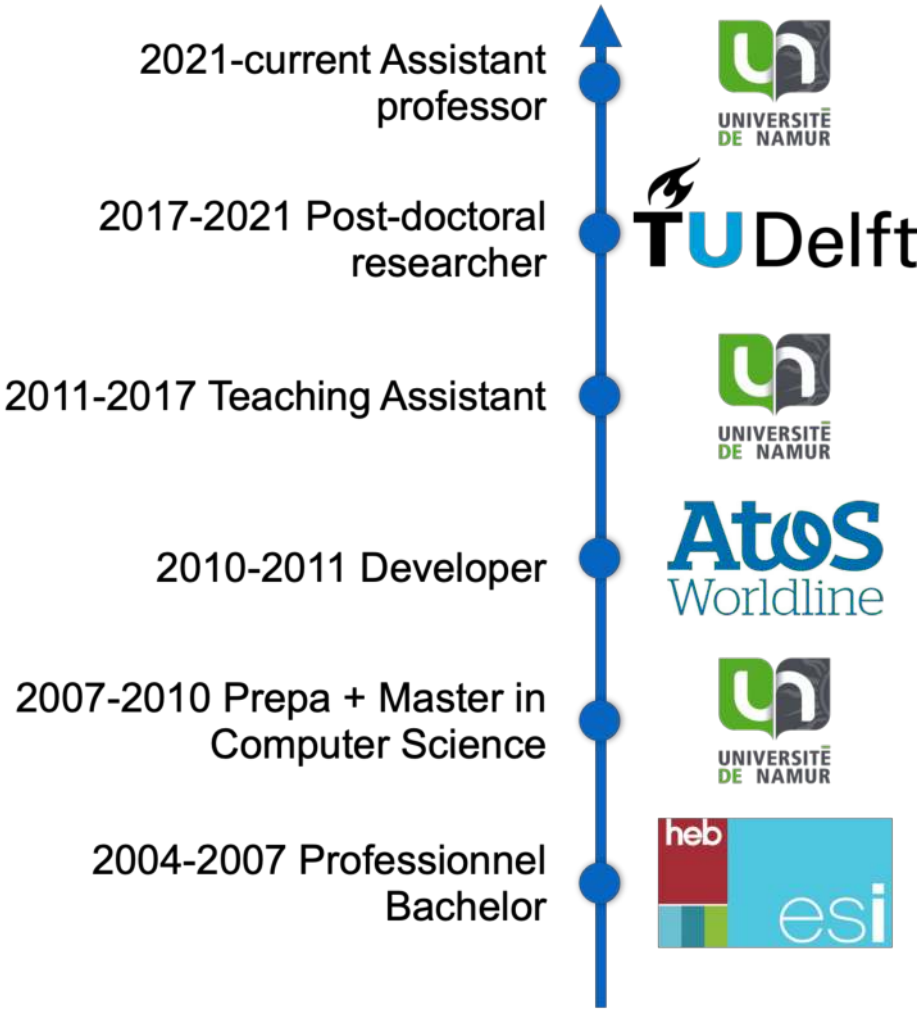
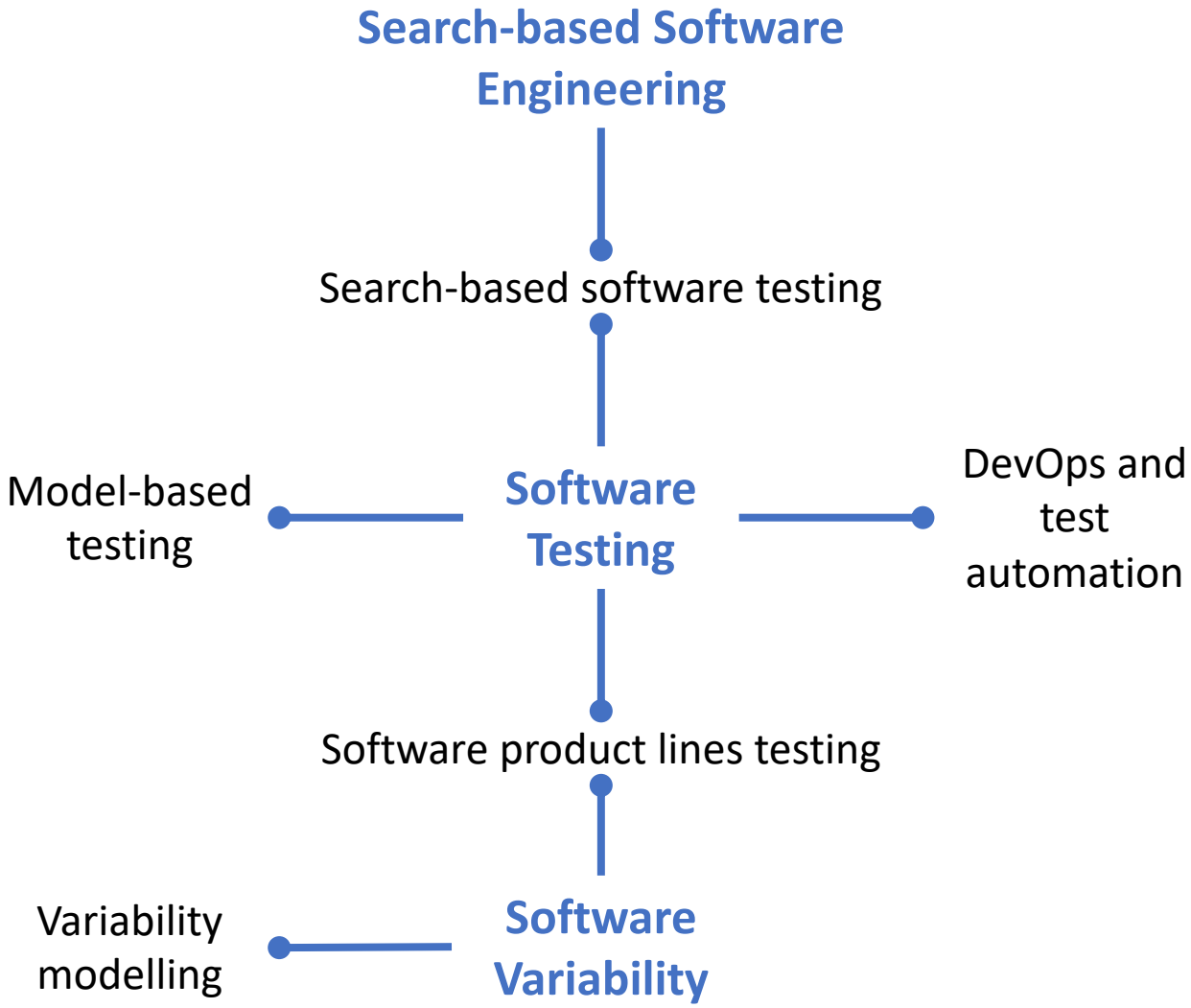
Background


2021-current Assistant professor
UNIVERSITÉ DE NAMUR
2017-2021 Post-doctoral researcher
TU Delft
2011-2017 Teaching Assistant
UNIVERSITÉ DE NAMUR
2010-2011 Developer
Atos Worldline
2007-2010 Prepa + Master in Computer Science
UNIVERSITÉ DE NAMUR
2004-2007 Professionnel Bachelor
heb esi


Search-based Software Engineering
Search-based software testing
Model-based testing
Software Testing
DevOps and test automation
Software product lines testing
Variability modelling
Software Variability

Software Normalization Assessment and Improvement Lab

Our team seeks to advance the state-of-the-art and practice of **software engineering** to improve software **reliability** by considering **quality** assessment and **testing** in both their **human** and **technical** contexts.

https://snail.info.unamur.be/

https://xdevroey.be/

What is *science*?

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What is *science*?

- "(knowledge from) the careful study of the structure and behaviour of the physical world, especially by watching, measuring, and doing experiments, and the development of theories to describe the results of these activities."
  (Cambridge Dictionary - https://dictionary.cambridge.org/dictionary/english/science)
- *"ensemble de connaissances, de travaux d'une valeur universelle, ayant pour objet l'étude de faits et de relations vérifiables, selon des méthodes déterminées (comme l'observation, l'expérience, ou les hypothèses et la déduction)."*
  (Le Robert - https://dictionnaire.lerobert.com/definition/science)
- "Science (from Latin *scientia* 'knowledge')[1] is a systematic enterprise that builds and organizes knowledge in the form of testable explanations and predictions about the world.[2][3][4]"
  (Wikipedia - https://en.wikipedia.org/wiki/Science)

What are the rules of science?

Scientific knowledge needs to be built on arguments and proofs — not on beliefs or authority.	Even the most pessimistic truth is better than fiction.	There is no democracy in science: all can be wrong, and a single person can be right.
Pluralism and open discussion, even speculation, need to be allowed.	Everything is a priori possible; the issue is only how probable it is.	Scientific proofs need to be built through research experiments that are repeatable and unambiguous.
The results of research experiments need to be reported in an unbiased, clear, concrete, coherent and logically-structured way.	Researchers need to be able to report on new knowledge in a popular, open-minded way.	Science neither has a final goal, nor final theories.

Hengl, T., Gould, M., 2006. The unofficial guide for authors (or how to produce research articles worth citing). EUR 22191 EN, 54 pp. Office for Official Publications of the European Communities, Luxemburg. ISBN: 92-79-01703-9 - https://op.europa.eu/s/uZYT

The world of science

Science

Researchers	Research group	Research organizations
System of science	Research publications	Commercial scientific companies

Hengl, T., Gould, M., 2006. The unofficial guide for authors (or how to produce research articles worth citing). EUR 22191 EN, 54 pp. Office for Official Publications of the European Communities, Luxemburg. ISBN: 92-79-01703-9 - https://op.europa.eu/s/uZYT

Researchers

- People who professionally conduct research experiments

Professors	Researchers and postdocs	PhD students	Research technicians
Tenured v.s. non tenured	Funded via project	Funded via project	Employee of the
Assistant Prof.	Funded via scholarship	Funded via scholarship	university
Associate Prof.	Tenured vs non tenured	Teaching assistant	
Full Prof.		PhD in a company	

Have a look: https://directory.unamur.be/entities/inf

Research groups

- Groups of researchers working on similar topics and meeting regularly at scientific meetings and workshops

	Local	International
Small	SNAIL TEAM@UNAMUR http://snail.info.unamur.be	ssbse https://ssbse.info
Large	NAD Namur Digital Institute https://nadi.unamur.be	SIGSOFT SPECIAL INTEREST GROUP ON SOFTWARE ENGINEERING https://www.sigsoft.org/

Research organizations

- Research societies, institutes and schools

Universities

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Research centers

cetic
Your Connection to ICT Research

Multitel
INNOVATION CENTRE

Private companies

Research and

Development

System of science

- National and international organizations with their members, structure, formal rules and evaluation criteria

National

International

Research publications

- Articles and books published by scientific publishers
- The purpose of a research paper is to communicate new, original knowledge (new concepts, new data), so that readers can apply, modify and extend that knowledge

https://researchportal.unamur.be

https://scholar.google.com

https://dblp.uni-trier.de

(Commercial) scientific companies

- Publishers, scientific information providers

Springer

WILEY

ELSEVIER

ACADEMIA

ResearchGate

IEEE
Advancing Technology for Humanity

acm

zenodo

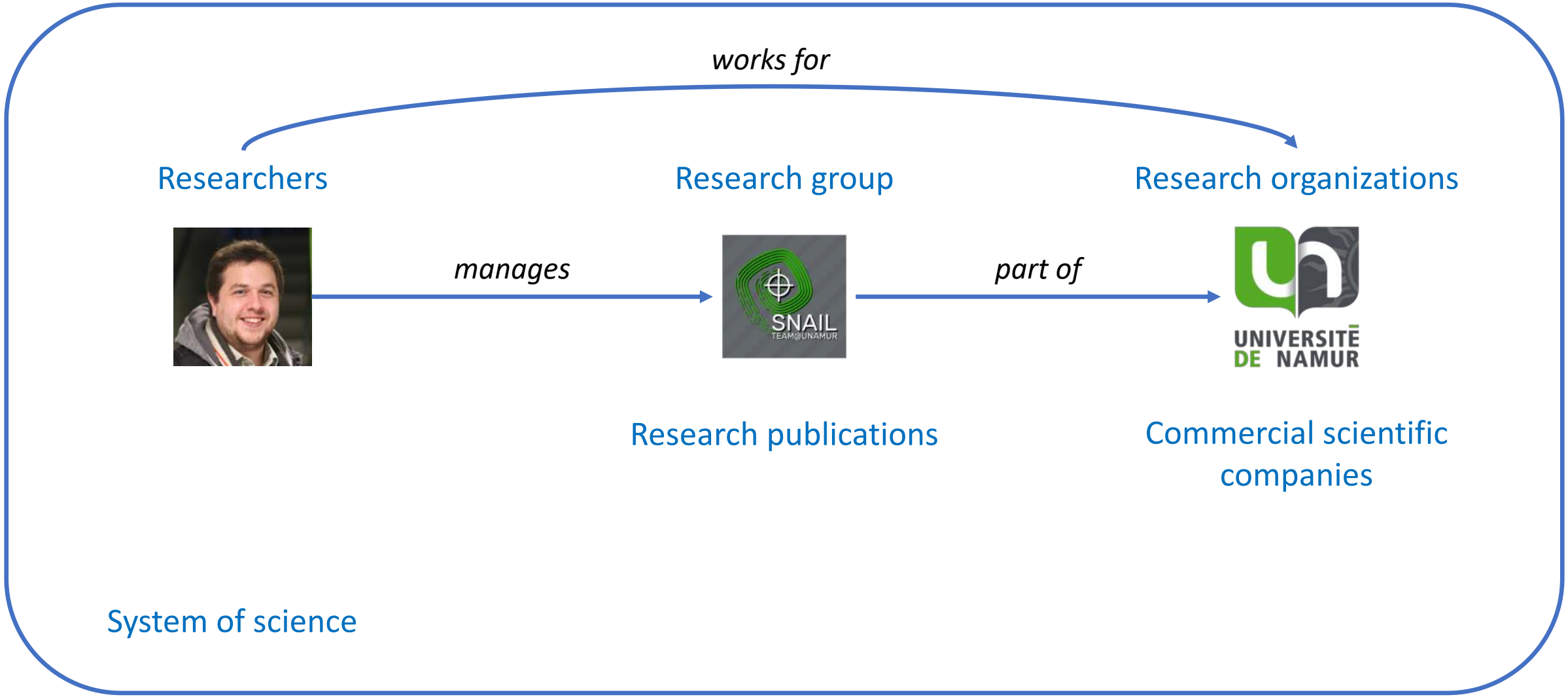
figshare

arXiv.org

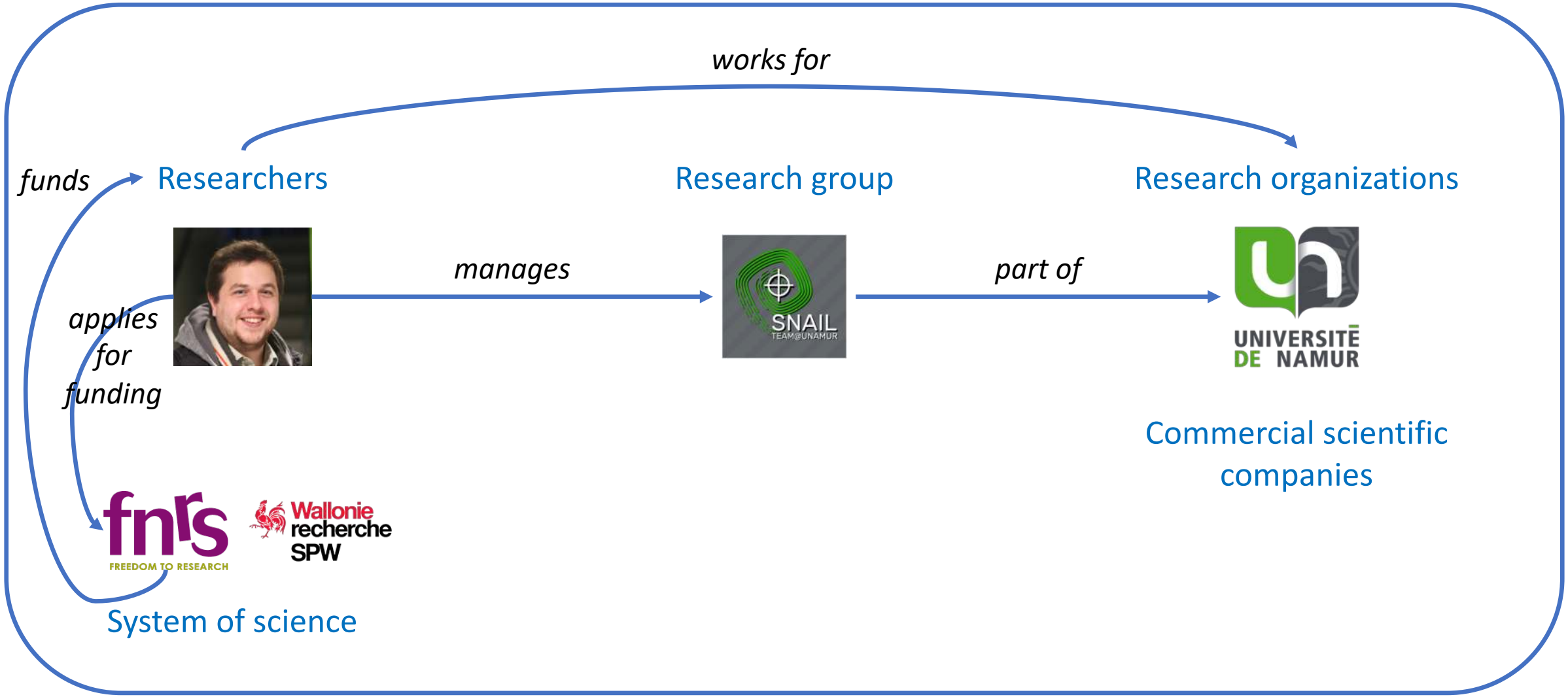
Sci-Hub

(Illegal)

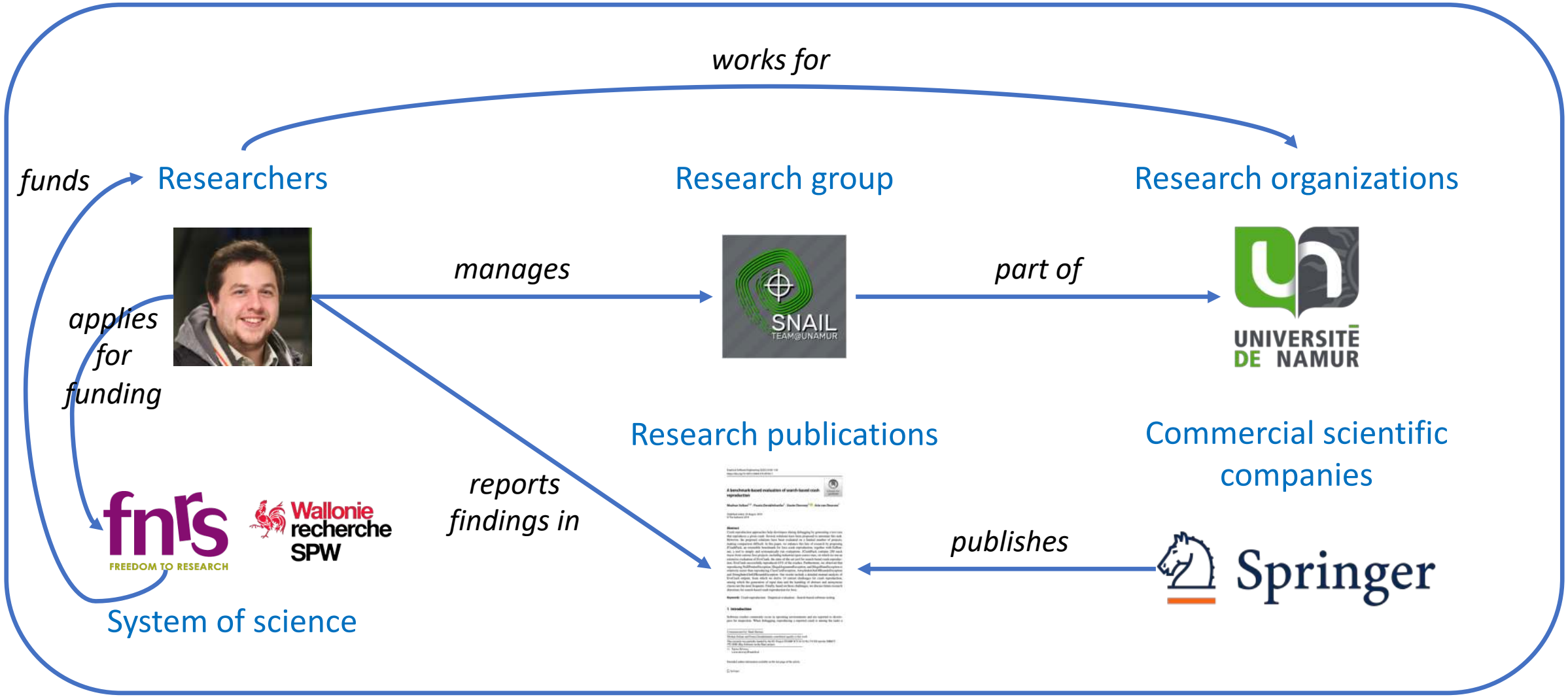
The world of science



The world of science



The world of science



Research and research method

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Ex.: Benchmarking Java Unit Test Generators

- Domain
- Analysis of related work
- Research question
- Methodology
- Results
- Limitations
- Research perspectives

JUGE: An Infrastructure for Benchmarking Java Unit Test Generators

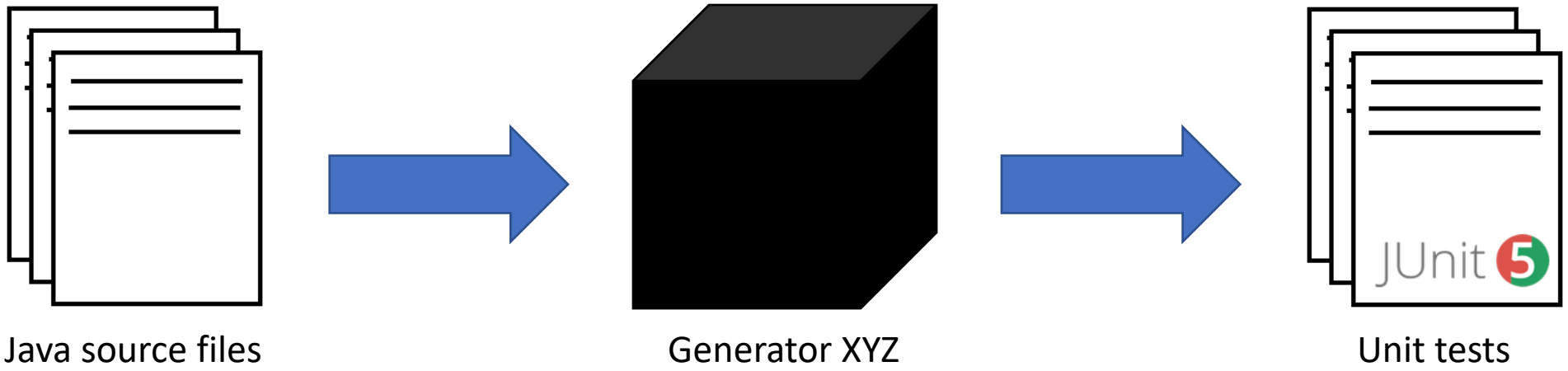
Xavier Devroey[1*], Alessio Gambi[2], Juan Pablo Galeotti[3], René Just[4], Fitsum Kifetew[5], Annibale Panichella[1], and Sebastiano Panichella[6],

[1] Delft University of Technology, Porthus 5, 2600 AA Delft, The Netherlands
[2] University of Passau, Innstrasse 41, 94032 Passau, Germany
[3] Universidad de Buenos Aires, Argentina
[4] University of Washington, Seattle, USA
[5] Fondazione Bruno Kessler, via Sommarive 18, 38123 Trento, Italy
[6] Zurich University of Applied Science, Steinberggasse 13, 8400 Winterthur, Switzerland

https://arxiv.org/abs/2106.07520

Domain

- Java programs
- White-box unit test cases generation



Analysis of related work

- Pseudo-random
  - C. Pacheco, et al., “Finding Errors in .NET with Feedback-directed Random Testing”, in ISSTA ‘08, 2008.
  - C. Pacheco and M. D. Ernst, “Randoop: Feedback-directed Random Testing for Java”, in OOPSLA ‘07 Companion, 2007.
  - C. Pacheco, et al., “Feedback-directed random test generation”, in ICSE ‘07, 2007.

- Evolutionary computing
  - Y. Lin, et al., “Graph-Based Seed Object Synthesis for Search-Based Unit Testing,” in ESEC/FSE ‘21, 2021.
  - S. Vogl, et al., “EVOSUITE at the SBST 2021 Tool Competition,” in SBST ‘21, 2021.
  - M. M. Almasi, et al., “An Industrial Evaluation of Unit Test Generation: Finding Real Faults in a Financial Application,” in SEIP@ICSE ‘17, 2017.
  - J. M. Rojas, et al., “A detailed investigation of the effectiveness of whole test suite generation,” in EMSE, 2016.

Randoop

Automatic unit test generation for Java

https://randoop.github.io/randoop/

EVOSUITE

Automatic Test Suite Generation for Java

https://www.evosuite.org

Research question

- RQ1. How does XYZ perform, compared to pseudo-random and evolutionary-based approaches?
  - RQ1.1. w.r.t. branch coverage?
  - RQ1.2. w.r.t. fault finding capabilities?
- RQ2. What factors influence the performance of XYZ?
  - RQ2.2. How does XYZ perform for different types of classes?
  - RQ2.2. How does XYZ perform for different configurations?
- RQ3. To what extend can developers include the tests generated by XYZ to existing test suites?
- …

Research question

- **RQ1. How does XYZ perform, compared to pseudo-random and evolutionary-based approaches?**
  - **RQ1.1. w.r.t. branch coverage?**
  - **RQ1.2. w.r.t. fault finding capabilities?**
- RQ2. What factors influence the performance of XYZ?
  - RQ2.2. How does XYZ perform for different types of classes?
  - RQ2.2. How does XYZ perform for different configurations?
- RQ3. To what extend can developers include the tests generated by XYZ to existing test suites?
- …

Methodology: Data Collection

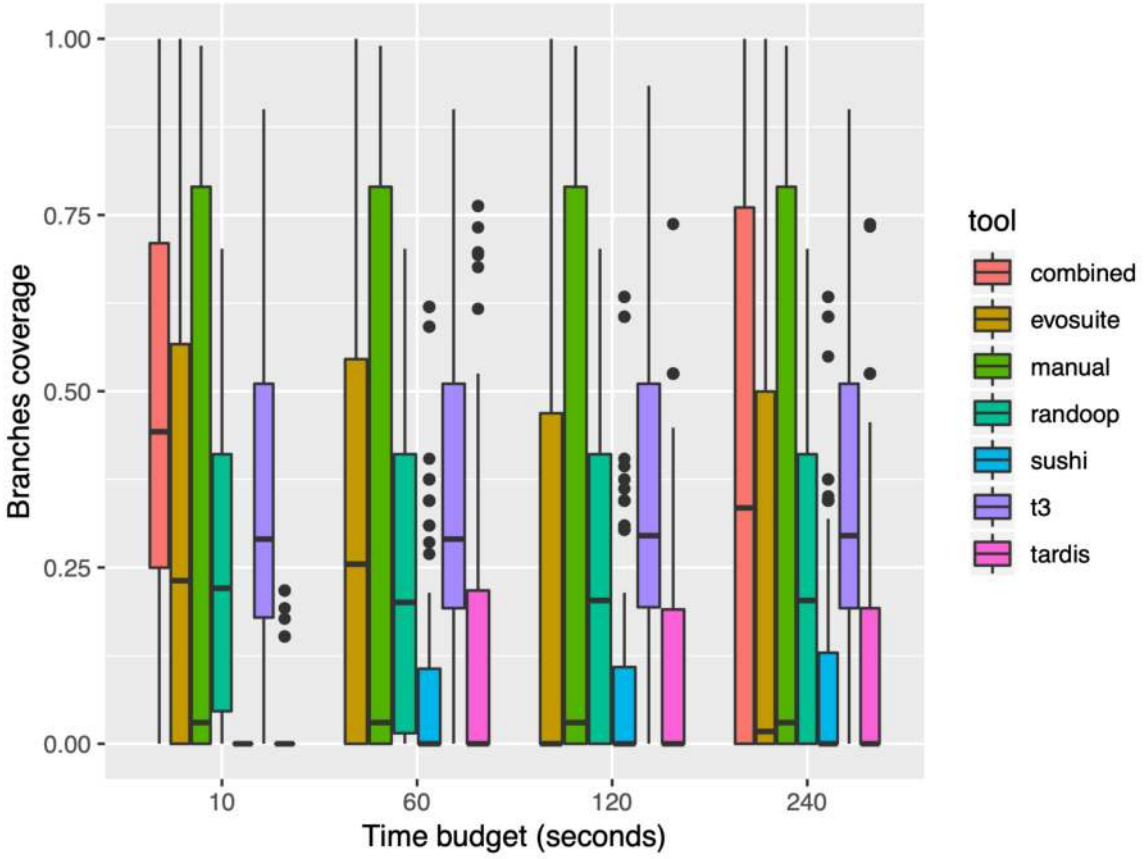
- Empirical evaluation
- Benchmarks
  - Defects4J
  - SF110
- How many classes under test?
- How many execution of XYZ on each class under test?
- How many configurations of XYZ?

Methodology: Data Analysis

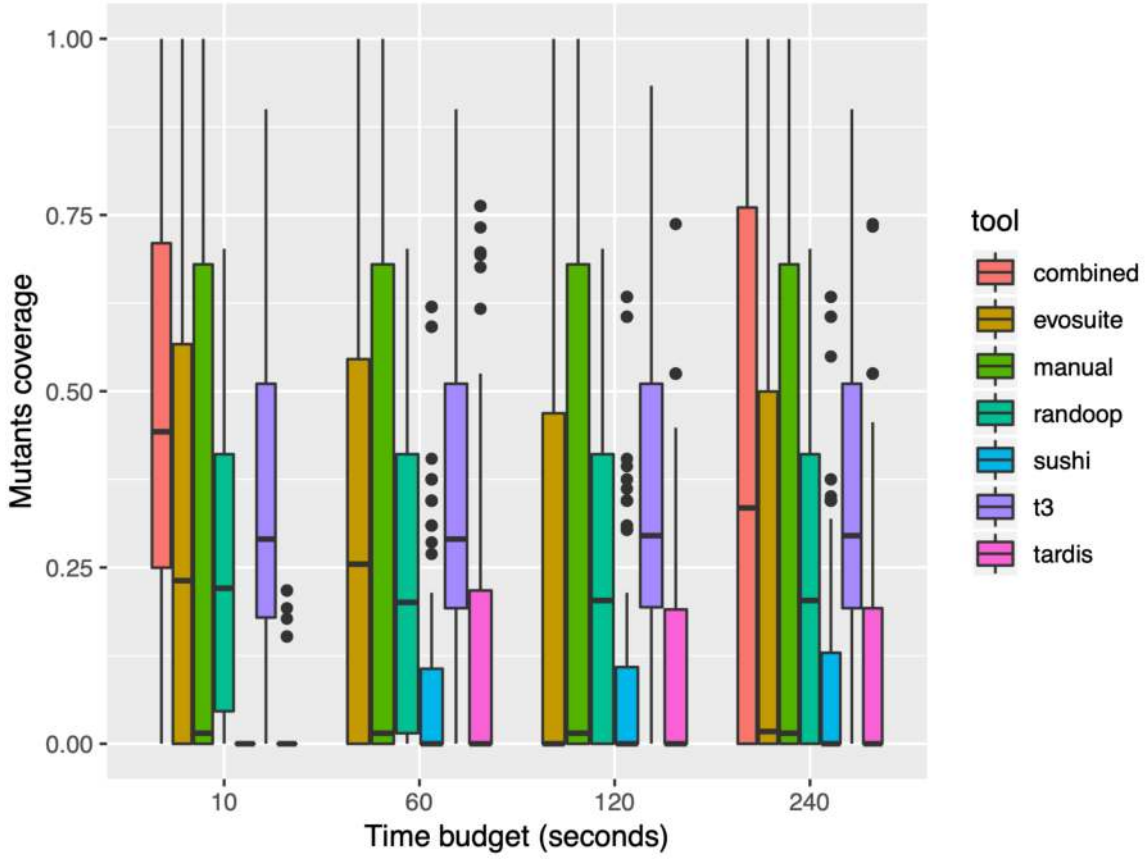
- Compare branch coverage and mutation score
- Visualization using boxplots
- Quantify the difference using *statistical tests* and *effect size measures*
  - Arcuri, A. and Briand, L. 2014. A Hitchhiker's guide to statistical tests for assessing randomized algorithms in software engineering. *Software Testing, Verification and Reliability*. 24, 3 (2014), 219–250. DOI: https://doi.org/10.1002/stvr.1486.

Results

Branch coverage



Mutation score



Kifetew, F., Devroey, X. and Rueda, U. 2019. Java Unit Testing Tool Competition - Seventh Round. 2019 IEEE/ACM 12th International Workshop on Search-Based Software Testing (SBST) (May 2019), 15–20. https://doi.org/10.1109/SBST.2019.00014

Results

Tool	Friedman's Test		Statistically better than (Conover's procedure)
	Rank	Score	
T3	1	2.30	Sushi, Tardis
EvoSuite	2	2.38	Sushi, Tardis
Randoop	3	2.51	Sushi, Tardis
Tardis	4	3.73	
Sushi	5	4.09	

Kifetew, F., Devroey, X. and Rueda, U. 2019. Java Unit Testing Tool Competition - Seventh Round. 2019 IEEE/ACM 12th International Workshop on Search-Based Software Testing (SBST) (May 2019), 15–20. https://doi.org/10.1109/SBST.2019.00014

Limitations

- The results must be interpreted with some limitations in mind
- Limited number of classes under test
  - How representative is it?
- Limited number of executions
  - Are the results due to luck?
- Limited number of configurations of the tools
  - Are there other configurations that would work better?
- Is the implementation of XYZ bug-free?
- Can the results be reproduced independently?

Research perspectives

- Combining XYZ with another existing approach
- Comparison with manually written tests
- Evaluation against other kinds of benchmarks
- …

Now it is your turn!

- **Domain**
- **Analysis of related work**
- **Research question**
- **Methodology**
- Results
- Limitations
- Research perspectives

REE, Article 81. Plagiat

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Définition

- §1er. Le plagiat consiste en l'utilisation, quelle qu'en soit l'ampleur, **le cas échéant par le biais d'une traduction**, des idées ou énonciations d'un tiers, fussent-elles paraphrasées, de même que l'utilisation de représentations graphiques d'un tiers, sans que leur source ne soit mentionnée explicitement et distinctement au sein de la prestation réalisée par l'étudiant.
- §2. Les enseignants peuvent contrôler la présence de plagiat dans les travaux et lors des évaluations.
  - **(C'est le cas pour les travaux remis en IDS)**

Conséquences

- §3. Lorsqu’ un enseignant constate le recours au plagiat dans le travail ou lors de l’évaluation d’un étudiant, il fait part de ce constat à l’étudiant et lui donne la possibilité de l’entendre à ce propos.
- Sur la base de l’ensemble des éléments ainsi récoltés, l’enseignant responsable de l’unité d’enseignement concernée peut attribuer la note de 0/20 pour l’évaluation concernée. Il peut, par ailleurs, **attribuer la note de 0/20 à l’unité d’enseignement concernée**, et ce, même si les faits se sont déroulés lors d’une évaluation portant sur une partie de l’unité d’enseignement concernée et dont le résultat constitue un élément de la note globale.

Conséquences

- Lorsque les faits de plagiat sont particulièrement graves, notamment lorsqu'il s'agit d'un **comportement répété ou pour lequel l'enseignant avait préalablement averti l'étudiant**, l'enseignant peut **en faire part au président et secrétaire du jury ainsi qu'au doyen et vice-doyen afin d'entamer la procédure applicable en cas de fraude aux évaluations** telle que décrite à l'article 77, §2, en vue de faire appliquer une des sanctions académiques telles que prévues à l'article 77, §3, différente que l'attribution de la note de 0/20.

A word about ChatGPT (and others)

What constitutes a plagiarism?

Plagiarism constitutes the unauthorized use of someone else's original work, including but not limited to: copying text directly, paraphrasing without proper citation, submitting work purchased or obtained from another source, using someone else's creative work (e.g., images, music, etc.) without permission, and fabrication of data or sources.

Is copy-pasting from ChatGPT a plagiarism?

Yes, copying text directly from ChatGPT and presenting it as one's own original work constitutes plagiarism. Proper citation or attribution should be given to the source of the information.

How should I acknowledge ChatGPT?

To acknowledge ChatGPT as a source, you can provide a reference or citation in a format appropriate for your assignment or use case. A common format is to include the following information:

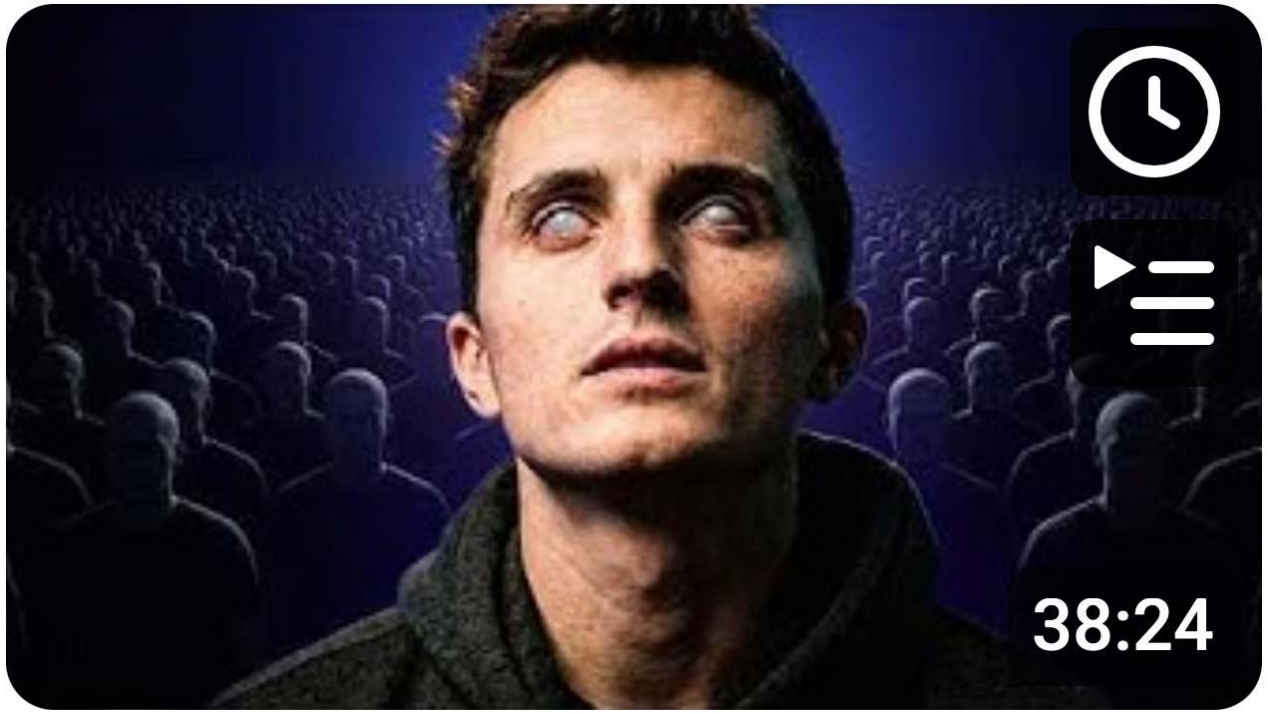
- Source: OpenAI's ChatGPT
- Date of access: (insert date of access)

Example citation: "Source: OpenAI's ChatGPT (accessed on [insert date])."

Note that citation styles vary depending on the field and the specific requirements of your assignment or use case. It is always best to consult with the appropriate style guide or your instructor for guidance on proper citation.

Source: OpenAI's ChatGPT (accessed on 31/01/2023).

Micode - La Fabrique à Idiots



La Fabrique à Idiots

2,6 M de vues • il y a 2 semaines

https://youtu.be/4xq6bVbS-Pw?si=qlsP-UuMNlfiO2rD

D1 Groupe et sujet

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D1 Groupe et sujet

- Au choix parmi les contextes proposés sur Webcampus
  - Ou proposer votre propre contexte
- Chaque contexte comporte plusieurs sujets (*topics of interest*)
  - **1 topic unique par groupe**
- Inscription sur Webcampus: groupes de 3
- Template du déliverable disponible sur Webcampus
- Annonce du choix du *contexte* + *topic of interest* via Teams

I DON'T LIKE ASKING HELP FROM PEOPLE, THEY ALWAYS PANIC.
WHAT DO YOU MEAN?
I JUST DON'T LIKE TO WORRY PEOPLE! THEY ALWAYS ASSUME IT'S AN ACUTE CRISIS.
AND THAT'S WHY I ALWAYS HOLD OFF ASKING FOR HELP FOR AS LONG AS I POSSIBLY CAN.
AND THERE ARE NO FLAWS IN THIS TACTIC.
NONE.
instagram: @green_fox_blue_fox
twitter: @foxes_in_love
tumblr: https://foxes-in-love.tumblr.com/